Gastroenterology & Hepatology in Critical Care – Chapter Summary

Bleeding • Liver Failure • Pancreatitis • Nutrition

ICU / EDIC Board Review • MCQ-Oriented

1. Upper GI Bleeding

- **Glasgow-Blatchford ≤ 1:** outpatient; **Rockall** after endoscopy
- **Restrictive transfusion Hb < 7** (< 8 cardiac) – ↓ rebleeding & mortality (Villanueva)
- **PPI IV** before endoscopy; **endoscopy < 24 h** (urgent < 12 h no benefit)
- **TXA:** no benefit, ↑ VTE (HALT-IT)
- **Forrest classification:** Ia spurting, Ib oozing, IIa visible vessel, IIb clot → endoscopic therapy
- **Failed endoscopy** → transarterial embolization → surgery
- **Erythromycin** before endoscopy improves visualization

2. Variceal Bleeding

- **Vasoactive:** terlipressin / octreotide 2–5 days
- **Antibiotic prophylaxis:** ceftriaxone 1 g/day × 7 days (↓ infection & mortality)
- **Band ligation within 12 h**
- **Pre-emptive TIPS within 72 h** (Child C 10–13, Child B > 7 with active bleeding)
- **Refractory:** balloon tamponade (Sengstaken–Blakemore / Minnesota) or esophageal stent → rescue TIPS
- **Gastric varices:** cyanoacrylate glue, BRTO
- Avoid over-transfusion (↑ portal pressure)

3. Lower GI Bleeding

- **Shock index > 1** → CTA first
- Colonoscopy after bowel prep in stable patients
- Causes: diverticular (most common), angiodysplasia, ischemic colitis, IBD, post-polypectomy
- **15% of hematochezia with instability is upper GI** → OGD

4. Acute Liver Failure

- **Definition:** coagulopathy (INR ≥ 1.5) + encephalopathy, no prior liver disease, < 26 weeks
- **Hyperacute** (< 7 days: paracetamol, viral A/E) → highest cerebral edema risk but best spontaneous survival
- **Causes:** paracetamol (West), viral hepatitis (East), drugs, autoimmune, Wilson, Budd–Chiari, AFLP / HELLP, ischemic
- **NAC** for all causes (benefit non-paracetamol early grades)
- **Ammonia > 150–200 µmol/L** → ↑ intracranial hypertension risk
- **Neuroprotection:** head up, Na 145–150 (HTS), avoid hypercapnia, mannitol, hypothermia 34–35 °C rescue, **early CRRT** (↓ ammonia)
- **Coagulopathy:** don't correct INR unless bleeding (prognostic marker, rebalanced hemostasis)

5. Transplant Criteria in ALF

King's College – Paracetamol	Non-paracetamol
pH < 7.30 after resuscitation, OR	INR > 6.5, OR 3 of:
All 3: INR > 6.5, Cr > 3.4 mg/dL, grade III/IV encephalopathy	Age < 10 or > 40; jaundice → encephalopathy > 7 days; INR > 3.5; bilirubin > 17.5 mg/dL; unfavorable etiology

- **Lactate > 3.5** (early) / > 3.0 (after fluids) adds prognostic value
- **Wilson's:** Coombs-negative hemolysis, low ALP:bilirubin ratio → transplant

6. Paracetamol Toxicity

- **Rumack–Matthew nomogram** (single acute ingestion, 4–24 h)
- **NAC** 21-h regimen (or SNAP 12-h): any ingestion > 150 mg/kg or level above line, unknown time, staggered
- **Fomepizole** adjunct in massive overdose; dialysis if level > 900–1000 mg/L with acidosis / coma
- Glutathione depletion → NAPQI → centrilobular necrosis

7. Acute-on-Chronic Liver Failure

- **EASL-CLIF:** organ failures (liver, kidney, brain, coagulation, circulation, lungs) – CLIF-C ACLF score
- Precipitants: infection, alcohol, GI bleed, drugs
- **SBP:** ascitic PMN > 250 → cefotaxime / ceftriaxone + **albumin 1.5 g/kg day 1, 1 g/kg day 3** (↓ HRS & mortality)
- **Hepatic encephalopathy:** lactulose + rifaximin; search precipitants
- **HRS-AKI:** terlipressin + albumin
- **Large-volume paracentesis:** 8 g albumin per L removed (> 5 L)
- **Hepatopulmonary** (orthodeoxia, ↑ shunt) vs **portopulmonary** (PAH)
- **Alcoholic hepatitis:** Maddrey ≥ 32 → prednisolone (STOPAH – 28-day benefit only); Lille at day 7

8. Acute Pancreatitis

- **Diagnosis (2 of 3):** pain, lipase / amylase > 3 × ULN, imaging
- **Revised Atlanta:** mild (no organ failure), moderate (transient OF < 48 h / local complications), severe (persistent OF > 48 h)
- **Fluids:** moderate aggressive – Ringer's lactate; aggressive → fluid overload (WATERFALL)
- **Early enteral nutrition** (NG = NJ) within 24–72 h
- **No prophylactic antibiotics**
- **ERCP < 24 h** only if cholangitis
- **Infected necrosis** (gas on CT, FNA): antibiotics (carbapenem) → **step-up approach**: percutaneous / endoscopic drainage → necrosectomy, delay ≥ 4 weeks (walled-off)
- **Abdominal compartment syndrome**, splenic vein thrombosis, pseudoaneurysm bleeding
- **Hypertriglyceridemia** (> 1000 mg/dL): insulin infusion, plasmapheresis

9. Mesenteric Ischemia

- **Pain out of proportion** to exam; ↑ lactate (late)
- **Types:** arterial embolism (AF – SMA), arterial thrombosis, **NOMI** (low flow, vasopressors), venous thrombosis
- **CTA** = diagnostic
- **Rx:** anticoagulation, revascularization (endovascular / open), resection, second-look laparotomy
- NOMI: intra-arterial papaverine, optimize perfusion

10. Other GI Emergencies

- **C. difficile:** oral vancomycin or **fidaxomicin**; fulminant → oral vanco + IV metronidazole ± rectal vanco; colectomy (or loop ileostomy) if toxic megacolon / perforation
- **Toxic megacolon:** colon > 6 cm + systemic toxicity
- **Ogilvie's** (pseudo-obstruction): cecum > 12 cm → neostigmine 2 mg IV (bradycardia – atropine), decompression
- **Acute cholecystitis:** antibiotics + cholecystectomy / percutaneous cholecystostomy
- **Acalculous cholecystitis:** critically ill, TPN → drainage
- **Cholangitis (Charcot / Reynolds pentad):** antibiotics + ERCP

11. Nutrition in ICU

- **Enteral within 24–48 h** if hemodynamically stable (small doses on low vasopressor ok)
- **Energy:** start trophic / ~70% target, 20–25 kcal/kg/day; full by day 3–7 (EFFORT Protein, NUTRIREA-3: lower is fine early)
- **Protein:** 1.2–1.3 g/kg/day (high protein no benefit; harm in AKI – EFFORT Protein)
- **PN:** if EN not feasible; delayed PN (day 8) better than early (EPaNIC)
- **Indirect calorimetry** = gold standard for energy
- **Gastric residual volume:** don't stop for < 500 mL; prokinetics (erythromycin, metoclopramide)
- **Refeeding syndrome:** ↓ PO_4 → restrict calories 48 h + replace
- **Glutamine:** harm (REDOXS)

12. Stress Ulcer Prophylaxis

- **Indications:** MV > 48 h + risk, coagulopathy, shock, history of GI bleed
- **PPI:** ↓ clinically important bleeding (REVISE), no mortality effect; possible ↑ pneumonia / C. diff
- SUP-ICU: pantoprazole no mortality difference

13. Abdominal Compartment Syndrome

- **IAH:** IAP ≥ 12 mmHg (bladder pressure, end-expiration, supine)

Grade	IAP
I	12–15
II	16–20
III	21–25
IV	> 25

- **ACS:** IAP > 20 + new organ dysfunction
- **APP = MAP – IAP** (target ≥ 60)
- **Rx:** NG / rectal decompression, sedation / NMB, drain ascites, negative fluid balance → **decompressive laparotomy**

14. High-Yield MCQ Pearls

- [x] UGIB: transfuse Hb < 7; TXA no benefit
- [x] Variceal bleed: terlipressin + ceftriaxone + banding < 12 h
- [x] Pre-emptive TIPS in Child C ≤ 13
- [x] ALF: NAC for all causes
- [x] Ammonia > 150–200 → cerebral edema risk
- [x] King's paracetamol: pH < 7.30
- [x] SBP: PMN > 250 + albumin 1.5 / 1 g/kg
- [x] Pancreatitis: no prophylactic antibiotics; avoid aggressive fluids
- [x] Infected necrosis: step-up, delay ≥ 4 weeks
- [x] Ogilvie's: neostigmine
- [x] EN within 24–48 h; delay PN (EPaNIC)
- [x] ACS: IAP > 20 + new organ failure

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